

29 Reactions at the Molecular Level

Why Are We Learning This?

Up to this point, we have treated reactions as mathematical relationships between reactants and products. We have written stoichiometric equations, performed mole balances, and used rate laws to design reactors. However, chemical reactions occur because molecules collide, bonds break, and new bonds form. To understand reactor behavior at a deeper level, we must distinguish between:

Overall Reactions and Actual Molecular Events

The stoichiometric equation tells us the overall change in composition, but it does not necessarily describe what individual molecules are doing.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Rate Laws
(Kinetics)



Equilibrium
(Reversible Reactions)



Equilibrium Conversion

(Thermodynamic Limitations)



One Reaction: Solving in terms of Reaction Progress



One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)



Catalytic Reactions



Numerical Methods
(Python)



Energy Balances



One Reaction: Non-Isothermal Reactor Design



Multiple Reactions



Reactor Safety



Challenging the Well-Mixed Assumption
(Residence Time Distributions)



► Challenging the Elementary Rate Law Assumption ◀
(Reaction Mechanisms)

Learning Objectives

By the end of this lecture, you should be able to:

- Define reaction extent and reaction rate.
- Explain the relationship between reaction rates and species generation rates.
- Distinguish between elementary and non-elementary reactions.
- Explain why not every stoichiometric equation corresponds to a molecular event.

- Define a reaction mechanism.
- Explain how overall reactions are built from elementary reaction steps.
- Recognize when an observed rate law does and does not imply an elementary reaction.

Big Idea

Observed Reaction



May Not Equal Molecular Reality



Need Reaction Mechanism

The reaction equation we write on paper is often a summary of many molecular events occurring behind the scenes.

Notes from <https://sites.engineering.ucsb.edu/~jbrow/chemreacfun/ch2/slides-stoi-2up.pdf>

Consider the reaction $\text{SiH}_4 + \text{SiH}_2 \rightleftharpoons \text{Si}_2\text{H}_6$, which is a step in a chemical vapor deposition process:

The reaction *extent* ξ keeps track of the number of times this reaction event occurs, i.e., it counts

The reaction *rate* is then

Where V is

What Does Reaction Extent Measure?

Reaction extent ξ keeps track of how many times the reaction event has occurred.

Physical Interpretation

Reaction Event Occurs



Reaction Extent Changes

If: $\Delta\xi > 0$

the reaction proceeds in the forward direction.

If: $\Delta\xi < 0$

the reaction proceeds in the reverse direction.

If: $\Delta\xi = 0$

there is no net reaction progress.

Why This Matters

Reaction extent provides a way to track reaction progress that is independent of which species is being monitored.

If $\Delta\xi$ is positive:

If $\Delta\xi$ is negative:

If $\Delta\xi$ is in equilibrium:

Note that for this reaction, it is easy to "link" the rate of reaction to the rate of generation or consumption of one of the reactant or product species, since

We can thus measure the reaction rate based on the change in species

However, for most reactions, it is not possible to measure the reaction rate. We hence must measure the rates of generation or production of individual species. This is why rates are often written

Why Do We Usually Measure Species Rates?

In theory, we would like to measure Reaction Rate directly. In practice, this is usually impossible.

Instead, we measure Rate of Consumption or Rate of Formation of individual species.

Example

For $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ we can observe:

NO disappearing

O₂ disappearing

NO₂ forming

and then use stoichiometry to infer the reaction rate.

Key Insight

Measure Species Rates



Use Stoichiometry



Determine Reaction Rate

An example of such a reaction is $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$, where

Consider the reaction $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$. Is it likely that this reaction could occur as a molecular event? Why or why not?

Let's answer this question by considering what must happen on the molecular level for the reaction to occur:

- $\text{O}=\text{O}$ bond must break

-

Consider a homogeneous gas phase reaction. The gas molecules:

For the reaction $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ to happen as one molecular event:

Why Is Molecularity Important?

The stoichiometric equation $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ appears to require three molecules to collide simultaneously.

Molecular View

For this reaction to occur in a single step: NO, NO, and O₂ would all need to collide simultaneously with the correct orientation and sufficient energy. This is extremely unlikely.

Consequence

Stoichiometric Equation $\neq$ Molecular Event

Many overall reactions cannot occur as written and must instead proceed through multiple elementary steps.

Consider a gas phase reaction occurring over a catalyst surface:

For the reaction $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ to happen as one molecular event:

So, to answer our question, no, this reaction is not likely to happen as a molecular event because

So, we've looked at two reactions so far, $\text{SiH}_4 + \text{SiH}_2 \rightleftharpoons \text{Si}_2\text{H}_6$ and $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$. What if I told you that the reaction rates for these reactions were: $r_{\text{Si}_2\text{H}_6} = k_{\text{Si}_2\text{H}_6} C_{\text{SiH}_4} C_{\text{SiH}_2}$ and $r_{\text{NO}_2} = k_{\text{NO}_2} C_{\text{NO}}^2 C_{\text{O}_2}$ but that the first reaction is elementary but the second one is not 🤖

Reactions tend to convey two different meanings:

1)

2) The actual molecular events that happen as the reaction proceeds

The first is a

the second is an

An elementary reaction is

What Is an Elementary Reaction?

An elementary reaction is a reaction that occurs exactly as written at the molecular level.

Examples

$A \rightarrow B$: Unimolecular

$A + B \rightarrow C$: Bimolecular

Both are plausible molecular events.

Why Not Three or More Molecules?

Simultaneous collisions involving three or more molecules are extremely improbable.

As a result: Most Elementary Reactions Are Unimolecular or Bimolecular

Elementary reactions are usually unimolecular or bimolecular because

If the rate of a non-elementary reaction happens to exhibit elementary kinetics, it is

I.e., the observed rate of $2NO + O_2 \rightarrow 2NO_2$ is $r_{NO_2} = k_{NO_2} C_{NO}^2 C_{O_2}$, this means that the

reaction $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ exhibits
however, the

reaction $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ is not an elementary reaction since

Elementary Kinetics Does Not Guarantee an Elementary Reaction

Suppose the observed rate law is: $r = k [\text{NO}]^2 [\text{O}_2]$

This rate law resembles what we might expect if: $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ were elementary.

However: Observed Rate Law $\neq$ Proof of Elementary Reaction

The same rate law can arise from a more complicated reaction mechanism.

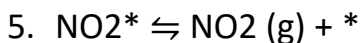
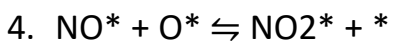
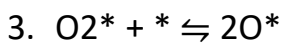
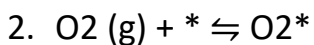
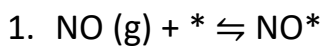
Key Insight

Elementary Kinetics May Be Observed Without An Elementary Reaction. Rate laws alone do not reveal the true molecular pathway.

Rather, this reaction likely occurs via a

which is comprised of

For example, consider the case where $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ occurs over a catalyst surface. One possible mechanism is:



What Is a Reaction Mechanism?

A reaction mechanism is a sequence of elementary reactions that collectively produce the observed overall reaction.

Example: Instead of $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ occurring in a single step, the reaction may proceed through: Adsorption, Surface Reactions, Desorption on a catalyst surface.

Physical Interpretation

Elementary Step 1

↓

Elementary Step 2

↓

Elementary Step 3

↓

...



Overall Reaction

The mechanism describes what molecules actually do. The stoichiometric equation only describes the net result.

For the overall reaction $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ is a

of reactions 1-5. The overall stoichiometry is reproduced by adding

Turning this into a vector:

Writing the stoichiometry matrix for the reaction mechanism:

We can then multiply s by ν to obtain

which is the overall stoichiometry statement for the reaction.

Why Use a Stoichiometry Matrix for Mechanisms?

Reaction mechanisms often contain many elementary steps and many intermediate species. The stoichiometry matrix provides a systematic way to: Track Species, Track Intermediates, and Recover Overall Stoichiometry from the individual elementary reactions.

Workflow

Elementary Reactions



Stoichiometry Matrix



Combine Steps



Overall Reaction

This is one of the reasons matrix methods are frequently used in reaction engineering.

Summary and Key Takeaways

Today we introduced chemical reactions from the molecular perspective.

Key Ideas

- ✓ Reaction extent tracks how many times a reaction event occurs.
- ✓ Reaction rates are usually inferred from species production or consumption rates.
- ✓ Overall stoichiometric equations do not necessarily represent molecular events.
- ✓ Elementary reactions occur exactly as written.
- ✓ Most elementary reactions are unimolecular or bimolecular.
- ✓ Observing elementary kinetics does not prove a reaction is elementary.
- ✓ Many overall reactions proceed through reaction mechanisms consisting of multiple elementary steps.
- ✓ Stoichiometry matrices provide a systematic way to analyze reaction mechanisms.

Most Important Idea

Overall Reaction



What We Observe

Reaction Mechanism



What Actually Happens

The stoichiometric equation describes the net chemical change, while the reaction mechanism describes the molecular events that produce that change. Understanding this distinction is the foundation for studying reaction mechanisms and catalytic kinetics.